

Part A – Graph Creation and Modification (30 pts)

Q1: Graph Creation (15 pts)

The graph shown in the assignment is an undirected graph on the ten vertices $V = \{0, 1, \dots, 9\}$. Reading the edges off the figure gives $|E| = 15$ edges:

$$E = \left\{ \begin{array}{l} (0, 1), (0, 4), (0, 5), (1, 2), (1, 6), \\ (2, 3), (2, 7), (3, 4), (3, 8), (4, 9), \\ (5, 7), (5, 8), (6, 8), (6, 9), (7, 9). \end{array} \right\}$$

Equivalently, as an adjacency list:

Node	Neighbours	Node	Neighbours
0	1, 4, 5	5	0, 7, 8
1	0, 2, 6	6	1, 8, 9
2	1, 3, 7	7	2, 5, 9
3	2, 4, 8	8	3, 5, 6
4	0, 3, 9	9	4, 6, 7

Construction. The graph was built in NetworkX by creating an empty undirected graph, adding the ten nodes $0, \dots, 9$, and then adding the fifteen edges listed above:

```
G = nx.Graph()
G.add_nodes_from(range(10))
G.add_edges_from(E)
```

Checks on the transcription. Every vertex has degree exactly 3, so the graph is 3-regular and the handshake lemma is satisfied: $\sum_v \deg(v) = 30 = 2|E| = 2 \cdot 15$. The graph is connected, has no triangles, and has girth 5. In fact these fifteen edges reproduce the *Petersen graph* (checked with NetworkX's `is_isomorphic` against `petersen_graph()`, which returns `True`), the standard 3-regular, girth-5 graph on ten vertices. Landing on the Petersen graph is a strong confirmation that the edge list was read off the drawing correctly.

One edge deserves comment, because the drawing in the assignment is misleading at that point: vertices 5, 4 and 8 are very nearly collinear, and the single edge (5, 8) is drawn passing straight across the disc of vertex 4. It therefore *looks* like a pair of edges (5, 4) and (4, 8), but it is one edge; if it were two, vertices 5, 4 and 8 would all have degree 4 and the graph would not be 3-regular. In Figure 1 the edge (5, 8) is drawn with a slight bow so that it visibly passes clear of vertex 4.

The visualization is shown in Figure 1. It uses the same vertex positions as the drawing in the assignment, so the two can be compared directly.

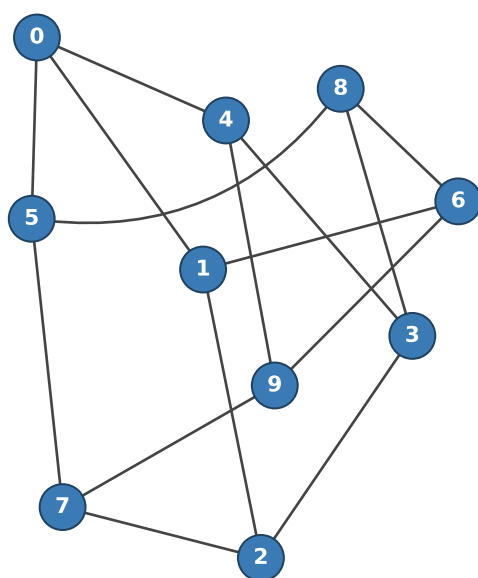


Figure 1: Q1: the undirected graph on 10 vertices and 15 edges, drawn with the same vertex placement as the figure in the assignment. The edge $(5, 8)$ is bowed so that it is visibly distinct from the vertex 4 it passes behind.

Q2: Graph Modification (15 pts)

Starting from the graph G of Q1 we remove, in order,

1. **node** 1, which also deletes its three incident edges $(0, 1)$, $(1, 2)$ and $(1, 6)$;
2. **edge** $(0, 4)$;
3. **edge** $(0, 5)$.

Five edges are removed in total, leaving the graph H on nine vertices with ten edges:

$$E(H) = \{(2, 3), (2, 7), (3, 4), (3, 8), (4, 9), (5, 7), (5, 8), (6, 8), (6, 9), (7, 9)\}.$$

	Before (Q1)	After (Q2)
Nodes	10	9
Edges	15	10
Connected components	1	2

Remaining components. The three edges at node 0 were exactly $(0, 1)$, $(0, 4)$ and $(0, 5)$, and all three are removed by the three operations above. Node 0 is therefore left isolated, and the resulting graph has **two** connected components:

Component	Vertices	Size
C_1	$\{0\}$	1
C_2	$\{2, 3, 4, 5, 6, 7, 8, 9\}$	8

C_2 contains all ten remaining edges and is connected; C_1 is the isolated vertex 0. The remaining graph is drawn in Figure 2, using the same vertex positions as Figure 1 so that the effect of the modification is directly visible; the isolated vertex 0 is drawn in orange.

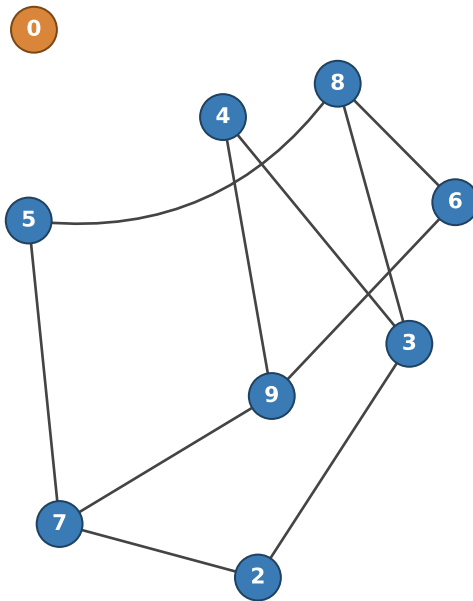


Figure 2: Q2: the graph after removing node 1 and the edges $(0,4)$ and $(0,5)$. Vertex positions are unchanged from Figure 1. The two connected components are the isolated vertex 0 (orange) and the remaining eight vertices.